BUSINESS REPORT

Hidden Gem Music

Cross-Regional Music Discovery Platform

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1. Executive Summary

Hidden Gem Music is a cross-regional music discovery platform built on real Spotify chart data spanning 73 countries and multiple years. The platform surfaces the "Discovery Gap" — the measurable lag between a song gaining traction in one country's charts and eventually crossing over into another market — and makes that gap visible and explorable through an interactive web and mobile application.

The project combines two Kaggle Spotify datasets totalling approximately 28.2 million chart rows, unified into a single SQL Server database and exposed through a .NET 9 API and a React Native frontend. The core analytical finding the platform demonstrates is that while music is globally accessible via streaming, recommendation algorithms and chart dynamics keep listening habits deeply regional. Songs can chart in the top 10 across a dozen countries while remaining completely absent from another market.

This report documents the data infrastructure, analytical methodology, key findings, and known limitations underpinning the Hidden Gem Music platform.

2. Problem Statement

Music discovery is regionally siloed. Despite global streaming access, Spotify's chart and recommendation systems reinforce local listening habits by surfacing locally popular content. There is no consumer-facing tool that clearly visualises this gap or helps users identify music they are statistically likely to enjoy based on cross-regional chart overlap.

The central question this platform is built to answer:

> *"Which globally trending artists and songs are missing from a given local market — and how do those patterns change over time?"*

Every data decision, schema design, stored procedure, KPI definition, and visualisation in this project exists to answer some version of that question. The Discovery Gap is not a feature — it is the entire point of the application.

3. Data Overview

The platform unifies two complementary Kaggle datasets that share subject matter but differ substantially in structure, geographic encoding, chart scope, and column naming.

3.1 Dataset 1 — Spotify Historical Charts (2017–2021)

Property	Value
Source	Kaggle — Spotify Historical Charts
Coverage	January 1, 2017 – December 31, 2021
File size	~3.5 GB
Raw rows (post-cleaning)	26,090,476
Ingested rows (post-dedup)	26,075,338
Chart types	Top 200 (77.6%) and Viral 50 (22.4%)
Countries	70 distinct regions (full country names)
Song identifier	Spotify URL (track ID parsed during ingestion)
Cleaning tool	Power BI — Power Query

3.2 Dataset 2 — Top Spotify Songs in 73 Countries (2023–2025)

Property	Value
Source	Kaggle — Top Spotify Songs in 73 Countries (Daily Updated)
Coverage	October 18, 2023 – June 11, 2025
File size	~497 MB
Raw rows (post-cleaning)	2,110,316
Chart type	Top 50 (daily)
Countries	73 distinct values including Global Top 50
Song identifier	Native Spotify ID (spotify_id)
Audio features	13 columns: danceability, energy, valence, tempo, and more
Cleaning tool	Power BI — Power Query (file exceeded Excel row limit)

3.3 Combined Dataset Scale

Source	ChartEntry Rows	Date Range
Dataset 1 (Historical Top 200 & Viral 50)	26,075,338	Jan 2017 – Dec 2021
Dataset 2 (Top 50, 73 Countries)	2,110,316	Oct 2023 – Jun 2025
Combined total	28,185,654	22-month gap confirmed

Note: A 22-month data gap exists between the two datasets (December 2021 – October 2023). No chart data is available for this period. All time-series visualisations in the application mark this gap explicitly, and it is disclosed to users on the application's Welcome Screen.

4. Data Quality & Cleaning

Both datasets were cleaned in Power BI using Power Query prior to SQL Server ingestion. Cleaning was performed in Power BI rather than Excel due to dataset sizes exceeding Excel's 1,048,576-row limit. The following documents the key issues identified and resolved for each dataset.

4.1 Dataset 1 — Key Quality Findings

Issue	Column(s)	Resolution
Missing values	region	NULL region values (Global Top 50 entries) filled with 'GLOBAL' to preserve as a valid, queryable category.
Expected NULLs	streams	All Viral 50 rows have NULL streams — expected and documented. Retained; filling with zero would be factually incorrect.
Incorrect data type	date	Stored as plain text. Converted to Date type in Power Query. No errors encountered.
Packed/delimited data	artist	Multiple artists in one comma-delimited string. Split into 4 positional columns (artist_primary through artist_4). Full relational split completed at SQL ingestion.
Duplicate rows	title + date + region + chart	Duplicates removed on composite key. Chart column included because the same song can legitimately appear on both Top 200 and Viral 50 on the same date.
Trend value discrepancy	trend	Source data uses SAME_POSITION, not SAME as documented. Ingestion script mapped correctly. ~3.3M rows would have been mis-normalised otherwise.

4.2 Dataset 2 — Key Quality Findings

Issue	Column(s)	Resolution
Missing values	country	NULL country values (Global Top 50) filled with 'GLOBAL'. Consistent with Dataset 1 treatment.
Movement column behaviour	daily_movement, weekly_movement	Zero NULLs found across 2.1M rows — unexpected. Confirmed Spotify uses integer 0 (not NULL) for first-appearance rows. Stored procedures use MIN(snapshot_date) rather than NULL checks to detect first appearances.
Incorrect data types	snapshot_date, album_release_date	Both stored as plain text. Converted to Date type in Power Query. No errors encountered.
Packed/delimited data	artists	Split into 7 positional columns (artist_primary through artist_7), reflecting the higher multi-artist crediting conventions of 2023–2025 vs. 2017–2021.
Duplicate rows	spotify_id + country + snapshot_date	Duplicates removed on composite key.

4.3 Schema Adjustments Required at Ingestion

Several column width constraints in the original schema were too narrow for actual data values encountered during BULK INSERT:

- Song title widened from NVARCHAR(255) to NVARCHAR(512) — max observed title length: 511 characters.
- Artist name widened from NVARCHAR(255) to NVARCHAR(1500) — max observed multi-artist string: 1,428 characters.
- Album name widened from NVARCHAR(255) to NVARCHAR(512) — album name exceeded 255 characters in Dataset 2.

Additionally, a SQL Server 2025 compatibility issue was encountered: ROWTERMINATOR escape sequences in BULK INSERT are treated as literal strings rather than byte values. Resolution: FORMAT = 'CSV' combined with ROWTERMINATOR = '0x0a' enables RFC 4180-compliant parsing and correctly interprets the hex terminator. This is documented and required for any future re-ingestion.

5. Database Architecture & Methodology

5.1 Schema Design — Star Schema

The database was initially designed in Third Normal Form (3NF). After both datasets were ingested and the workload characteristics became clear, the schema was migrated to a star schema dimensional model. The decision was documented in a formal Architecture Decision Record (ADR).

The migration rationale:

- The database is exclusively historical and analytical — data is written once at ingestion and never updated. 3NF's protections against write anomalies are irrelevant to this workload.

- Every operation the application performs is a read: aggregation, cross-country comparison, trend analysis, and Hidden Gem scoring. Star schema is specifically designed to optimise this kind of repeated analytical read.
- 3NF required a minimum of four joined tables (Song → ArtistSong → Artist → Album) for every query that returned song and artist data. Across 28 million ChartEntry rows, this join overhead compounds significantly.

Resulting schema structure:

Table	Purpose
FACT_ChartEntry	Central fact table. 28.2M rows. One row per song, per country, per chart date. All stored procedures ultimately derive from this table.
DIM_Song	Flat song dimension. Includes all audio features as nullable columns (Dataset 2 only). No separate AudioFeatures table.
DIM_Artist	One row per unique artist name.
Bridge_SongArtist	Resolves many-to-many between DIM_Song and DIM_Artist. artist_order records position in source data string.
Country	Bridges Dataset 1 full country names with Dataset 2 ISO codes. Includes latitude/longitude for map rendering.
ChartType	Three chart types: Top 200, Viral 50, Top 50.

5.2 Pre-Computed Summary Tables

At 28 million rows, any stored procedure scanning FACT_ChartEntry at request time is too slow for real-time API use. All heavy computation runs at population time and is stored in pre-computed summary tables. The API queries only these lightweight tables at runtime.

Summary Table	What It Pre-Computes
HiddenGems	For every country and year: songs charting in N+ other countries but absent from this country. Most expensive derivation — computed once, read at request time.
CountryYearStats	One row per country per year: total charted, shared count, unique count, overlap percentage.
SongCountryPresence	Per song per year: count of distinct countries it charted in. Foundation of Hidden Gems logic.
GlobalOverlapByYear	Year-by-year global overlap percentage and average countries per charting song. Includes is_gap flag for the 22-month data gap.
DiscoveryGapByDay	Per song: origin country, spread country, days between first chart appearances. Feeds KPI 2 and the gap distribution histogram.
IsolationScoreByCountry	Per country: local song count, shared song count, isolation percentage. Minimum 100 chart entry threshold applied.
PeakReachBySong	Per song: maximum simultaneous country count across all dates. Feeds KPI 4.

Summary Table	What It Pre-Computes
TopSongByCountryYear	Pre-computed top song per country per year. Enables fast Discovery Map load without scanning the full fact table.

5.3 API & Backend Layer

All database logic is encapsulated in stored procedures. The .NET 9 API layer (using Controller/Repository Pattern with DTOs) calls stored procedures and returns clean, structured JSON to the frontend. The API never receives or returns raw ChartEntry rows — only pre-aggregated results.

This architecture keeps heavy computation server-side and ensures the React Native frontend remains lightweight, regardless of the underlying dataset size.

6. Key Analytical Findings — Dashboard KPIs

The Global Overlap Dashboard presents four headline KPIs that directly demonstrate the Discovery Gap. Each is calculated via stored procedure against pre-computed summary tables and filterable by date range.

#	KPI	What It Measures	Why It Matters
1	Global Overlap Rate	Of all songs that charted across any country, what percentage appeared in 2+ countries' charts?	The headline number. A low overlap rate is direct evidence that music discovery is regionally siloed.
2	Average Discovery Gap (Days)	When a song first charts anywhere, how many days on average before it charts in a second country?	Makes the abstract problem concrete. Quantifies the literal lag in cross-regional discovery. Median is surfaced alongside mean — outliers from the cross-dataset gap skew the mean significantly.
3	Most Globally Isolated Region	Which country has the lowest overlap with global chart trends — the highest % of locally unique songs?	Gives the geographic dimension a headline. Identifies where the Discovery Gap is largest and provides a point of interest for globe exploration.
4	Peak Cross-Regional Reach	The maximum number of countries a single song simultaneously charted in at any point in the dataset.	The ceiling number. Makes the Discovery Gap meaningful by contrast — if peak reach is 60+ countries and the average is 3, that gap is the story.

KPI 2 note: songs that only ever charted in a single country are excluded (no second country date exists). The days_to_spread floor is set to > 1 to exclude songs released simultaneously across markets, which would otherwise produce artificially small gap values.

7. Known Limitations

7.1 The 22-Month Data Gap

The most significant structural limitation is the 22-month gap between Dataset 1 (ends December 2021) and Dataset 2 (begins October 2023). Songs that first charted in Dataset 1 and reappeared in Dataset 2 will show artificially inflated Discovery Gap values — the calculation sees a gap of hundreds or thousands of days rather than the true cross-regional spread time.

Mitigation: the gap is marked visually in all time-series charts, disclosed on the application Welcome Screen, and flagged in stored procedure logic via an is_gap BIT column. Median is surfaced alongside mean for KPI 2 specifically because outlier gap values from cross-dataset songs skew the mean significantly.

7.2 Cross-Dataset Comparability

Dataset 1 measures popularity via raw stream counts. Dataset 2 uses Spotify's proprietary 0–100 popularity score, whose methodology is not publicly disclosed. These metrics are never compared directly. Both are stored in separate nullable columns on FACT_ChartEntry and are clearly labelled in any visualisation that surfaces them.

7.3 Audio Features — Dataset 2 Only

All 13 audio feature columns (danceability, energy, valence, tempo, etc.) exist only in Dataset 2. Songs that appeared only in Dataset 1 carry NULL values for all audio feature fields. Any future feature using audio data must handle NULLs as expected and cannot apply to the 2017–2021 portion of the dataset.

7.4 Artist Name Deduplication

SQL Server cannot create unique indexes on NVARCHAR(MAX) columns. The DIM_Artist table enforces artist name uniqueness during ingestion via NOT EXISTS logic rather than a database constraint. Minor formatting variations in artist names across datasets may produce duplicate DIM_Artist rows. This is a known limitation of the MAX column type and is acceptable for a fixed, non-updating dataset.

7.5 Mobile Chart Rendering

The Recharts charting library used for web chart components is not compatible with React Native mobile. Mobile screens serve static screenshot-based chart representations as an intentional deferral, documented as a known platform limitation. A full mobile-native charting implementation would require switching to a different library (e.g. Victory Native) and is scoped as a future enhancement.

8. Conclusions

Hidden Gem Music demonstrates that the Discovery Gap is real, measurable, and quantifiable at scale. Using 28 million chart rows across 73 countries and eight years of data, the platform makes visible what streaming algorithms obscure: a song can be a chart phenomenon in multiple countries while remaining completely absent from markets where listeners would likely enjoy it.

The analytical infrastructure built for this project — a star schema database, eight pre-computed summary tables, 17+ stored procedures, and a .NET 9 API — reflects production-grade data engineering choices driven by the dataset's scale and the application's read-heavy workload. Every architectural decision, from the star schema migration to the pre-computation strategy, was made to keep the platform fast and analytically accurate regardless of the underlying data volume.

The four dashboard KPIs provide the clearest summary of what the data shows: the Global Overlap Rate quantifies how siloed music markets actually are; the Average Discovery Gap puts a number of days on the lag; the Most Isolated Region identifies where the gap is largest; and the Peak Cross-Regional Reach demonstrates what maximum global penetration looks like — making the gap between that ceiling and the average deeply meaningful.

Future enhancements would include full audio feature analysis across both dataset periods (pending additional data sourcing), mobile-native interactive charts, and a live data feed to replace the historical snapshot model.

9. Supporting Documentation

The following documents support the findings and decisions in this report and are available in the project BDA Reference folder:

Document	Description
Data Quality Report — Dataset 1	Quality assessment and cleaning decisions for the Historical Top 200 & Viral 50 dataset.
Data Quality Report — Dataset 2	Quality assessment and cleaning decisions for the Top 50, 73 Countries dataset.
EDA — Dataset 1 (Pre-Ingestion)	Exploratory data analysis conducted in Power BI prior to SQL ingestion.
EDA — Dataset 1 (Post-Ingestion)	Post-ingestion validation results and confirmed row counts for Dataset 1.
EDA — Dataset 2 (Pre-Ingestion)	Exploratory data analysis for Dataset 2, including the movement column finding.
EDA — Dataset 2 (Post-Ingestion)	Post-ingestion validation and combined database state confirmation.
Database Schema ADR	Architecture Decision Record documenting the original 3NF schema design and rationale.

Document	Description
Star Schema Migration ADR	ADR documenting the migration from 3NF to star schema, including full trade-off analysis.
KPI Definitions	Detailed formula definitions, SP signatures, and display guidance for all four dashboard KPIs.
Database Creation Script	Full SQL Server creation script for the HiddenGemMusic database including schema and lookup data.
Project Proposal	Original capstone proposal documenting scope, technology stack, milestones, and evaluation criteria.